20261117 Quiz

for (MATH230) Probability & Statistics

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Exercise 8.20. An unbiased die is tossed independently 72 times. If X denotes one outcome, its probability mass function is

$$\mathbb{P}(X = x) = \begin{cases} 1/6, & x = 1, 2, 3, 4, 5, 6, \\ 0, & \text{otherwise.} \end{cases}$$

Find the probability that the sample mean is greater than 3.6 but less than 3.9, when means are recorded to the nearest tenth.

Solution. For a single throw, symmetry and the second moment give

$$\mu = \frac{1+2+3+4+5+6}{6} = \frac{7}{2}, \qquad \sigma^2 = \frac{1^2+2^2+\cdots+6^2}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}.$$

Thus $\sigma_{\bar{X}} = \sqrt{35/(12 \cdot 72)} \approx 0.20127$. Since the reported mean is rounded to the nearest tenth, the requested reported values correspond to $3.65 \le \bar{X} < 3.85$. By the Central Limit Theorem,

$$\begin{aligned} \mathbb{P}(3.65 \le \bar{X} < 3.85) &\approx \Phi\left(\frac{3.85 - 3.5}{0.20127}\right) - \Phi\left(\frac{3.65 - 3.5}{0.20127}\right) \\ &= \Phi(1.739) - \Phi(0.745) \approx \boxed{0.1870}. \end{aligned}$$

□

Exercise 8.22. The heights of 1000 students are approximately normally distributed with mean 174.5 cm and standard deviation 6.9 cm. Suppose 200 random samples, each of size 25, are drawn, and each sample mean is recorded to the nearest tenth of a centimeter. Determine

(a) the mean and standard deviation of the sampling distribution of $\bar{X}$;
(b) the expected number of recorded sample means between 172.5 and 175.8 cm, inclusive;
(c) the expected number of recorded sample means below 172.0 cm.

Solution. As in the book's sampling-distribution approximation, use $\bar{X} \approx \mathcal{N}(174.5, 6.9^2/25)$, so

(a) The mean and standard error are

$$\boxed{\mu_{\bar{X}} = 174.5 \text{ cm}}, \qquad \boxed{\sigma_{\bar{X}} = \frac{6.9}{\sqrt{25}} = 1.38 \text{ cm}}.$$

(b) The inclusive reported interval corresponds to $172.45 \le \bar{X} < 175.85$. Hence the expected count among 200 sample means is

$$\begin{aligned} 200\mathbb{P}(172.45 \le \bar{X} < 175.85) &\approx 200\left[\Phi\left(\frac{175.85 - 174.5}{1.38}\right) - \Phi\left(\frac{172.45 - 174.5}{1.38}\right)\right] \\ &\approx \boxed{153.5 \text{ (about 153–154 means)}}. \end{aligned}$$

(c) A reported mean below 172.0 corresponds to $\bar{X} < 171.95$. Thus

$$200\mathbb{P}(\bar{X} < 171.95) \approx 200\Phi\left(\frac{171.95 - 174.5}{1.38}\right) \approx \boxed{6.46 \text{ (about 6–7 means)}}.$$

□

Exercise 8.23. The number X of cherries in a cherry puff has the probability distribution

x	4	5	6	7
$\mathbb{P}(X = x)$	0.2	0.4	0.3	0.1

(a) Find the mean μ and variance σ^2 of X.
(b) Find the mean and variance of $\bar{X}$ for independent random samples of 36 cherry puffs.
(c) Find the probability that the average number of cherries in 36 puffs is less than 5.5.

Solution.

(a) Direct calculation yields

$$\begin{aligned} \mu &= 4(0.2) + 5(0.4) + 6(0.3) + 7(0.1) = \boxed{5.3}, \\ \mathbb{E}[X^2] &= 16(0.2) + 25(0.4) + 36(0.3) + 49(0.1) = 28.9, \\ \sigma^2 &= 28.9 - 5.3^2 = \boxed{0.81}. \end{aligned}$$

(b) For a sample of size 36,

$$\boxed{\mathbb{E}[\bar{X}] = 5.3}, \qquad \boxed{\operatorname{Var}(\bar{X}) = \frac{0.81}{36} = 0.0225}.$$

(c) The standard error is $\sqrt{0.0225} = 0.15$. By the Central Limit Theorem,

$$\mathbb{P}(\bar{X} < 5.5) \approx \Phi\left(\frac{5.5 - 5.3}{0.15}\right) = \Phi(1.3333) \approx \boxed{0.9088}.$$

□

Exercise 8.24. A machine produces electrical resistors with mean resistance $40\,\Omega$ and standard deviation $2\,\Omega$. What is the probability that a random sample of 36 resistors has a combined resistance greater than $1458\,\Omega$?

Solution. Let $W = X_1 + \cdots + X_{36}$ be the total resistance. Independence gives

$$\mathbb{E}[W] = 36(40) = 1440\,\Omega, \qquad \operatorname{Var}(W) = 36(2^2) = 144\,\Omega^2.$$

By the Central Limit Theorem, $W \approx \mathcal{N}(1440, 144)$, and therefore

$$\mathbb{P}(W > 1458) \approx 1 - \Phi\left(\frac{1458 - 1440}{12}\right) = 1 - \Phi(1.5) \approx \boxed{0.0668}.$$

□

Exercise 8.27. The impurity content (grams per gram of output) in a chemical process has known population standard deviation 0.1. It is conjectured that the population mean is $\mu = 0.20$. In 50 laboratory runs the observed mean impurity content is $\bar{x} = 0.23$. Use the Central Limit Theorem to comment on the conjecture.

Solution. Assuming $\mu = 0.20$, the standard error of the sample mean is $0.1/\sqrt{50}$. Thus

$$z = \frac{0.23 - 0.20}{0.1/\sqrt{50}} \approx 2.1213,$$

and the chance of obtaining a sample mean at least as large as observed is

$$\mathbb{P}(\bar{X} \geq 0.23) \approx 1 - \Phi(2.1213) \approx \boxed{0.0169}.$$

Such a large mean would occur in roughly 1.7% of repeated samples under the conjecture. The observation therefore casts doubt on $\mu = 0.20$, though it does not prove the conjecture false. □

Exercise 8.28. Independent random samples of sizes 30 and 42 are drawn from adults in cities A and B, respectively. The population heights have means 160 and 158 cm and standard deviations 9 and 10 cm, respectively. Find the probability that the difference $\bar{X}_A - \bar{X}_B$, recorded to the nearest tenth of a centimeter, is at least 2.6 but less than 4.8 cm.

Solution. For $D = \bar{X}_A - \bar{X}_B$, independence gives

$$\mathbb{E}[D] = 160 - 158 = 2, \qquad \mathrm{Var}(D) = \frac{9^2}{30} + \frac{10^2}{42} \approx 5.08095,$$

so $\sigma_D \approx 2.25410$. The stated recorded values correspond to the unrounded interval $2.55 \leq D < 4.75$. With the normal approximation to the difference of means,

$$\begin{aligned}\mathbb{P}(2.55 \leq D < 4.75) &\approx \Phi\left(\frac{4.75 - 2}{2.25410}\right) - \Phi\left(\frac{2.55 - 2}{2.25410}\right) \\ &= \Phi(1.220) - \Phi(0.244) \approx \boxed{0.2924}.\end{aligned}$$

□

Exercise 8.29. One breed of one-week-old chicks has mean weight 120 g and standard deviation 12 g; a second breed has mean 92 g and standard deviation 8 g. Independent samples contain 46 and 84 chicks, respectively. Find the probability that the first sample mean exceeds the second by at most 20 g. Assume the means can be measured to any degree of accuracy.

Solution. For $D = \bar{X}_1 - \bar{X}_2$,

$$\mathbb{E}[D] = 28, \qquad \mathrm{Var}(D) = \frac{12^2}{46} + \frac{8^2}{84} \approx 3.89234, \quad \sigma_D \approx 1.97290.$$

The two sample sizes allow a normal approximation, so

$$\mathbb{P}(D \leq 20) \approx \Phi\left(\frac{20 - 28}{1.97290}\right) = \Phi(-4.055) \approx \boxed{0.0000251}.$$

The event is extremely unlikely under these population parameters. □

Exercise 8.30. Students' scores on a competitive examination have population mean 148 and standard deviation 24. Two independent random groups of sizes 42 and 64 are selected. Assuming sample means can be measured to any degree of accuracy, find the probability that their means differ by

(a) more than 12 points;
(b) between 4 and 10 points.

Solution. For $D = \bar{X}_1 - \bar{X}_2$, the mean difference is 0 and

$$\mathrm{Var}(D) = \frac{24^2}{42} + \frac{24^2}{64}, \qquad \sigma_D \approx 4.76595.$$

Using the normal approximation for the two independent sample means,

(a) The question concerns the absolute difference:

$$\mathbb{P}(|D| > 12) \approx 2\left[1 - \Phi\left(\frac{12}{4.76595}\right)\right] \approx \boxed{0.0118}.$$

(b) By symmetry,

$$\begin{aligned}\mathbb{P}(4 < |D| < 10) &\approx 2\left[\Phi\left(\frac{10}{4.76595}\right) - \Phi\left(\frac{4}{4.76595}\right)\right] \\ &\approx \boxed{0.3654}.\end{aligned}$$

□

Exercise 8.31. Two types of paint have population drying-time standard deviations of 1.0 hour each. In independent samples of 18 specimens per type, the observed difference of sample mean drying times is $\bar{x}_A - \bar{x}_B = 1.0$ hour. Assume a normal approximation for the difference of sample means.

(a) Is this a reasonable result if the population means are equal? Use the probability from Case Study 8.2.
(b) If the experiment is repeated 10 000 times under $\mu_A = \mu_B$, how many experiments are expected to give $\bar{x}_A - \bar{x}_B \geq 1.0$ hour?

Solution. Under $\mu_A = \mu_B$, the difference $D = \bar{X}_A - \bar{X}_B$ has

$$\mathbb{E}[D] = 0, \qquad \sigma_D = \sqrt{\frac{1}{18} + \frac{1}{18}} = \frac{1}{3} \text{ hour}.$$

(a) A difference of at least one hour has probability

$$\mathbb{P}(D \geq 1) \approx 1 - \Phi\left(\frac{1}{1/3}\right) = 1 - \Phi(3) \approx \boxed{0.00135}.$$

It is very rare under equal population means, so the observation is evidence against equality.

(b) The expected number of such experiments is

$$10\,000(0.00135) \approx \boxed{13.5 \text{ (about 13–14 experiments)}}.$$

□

Exercise 8.32. Two cereal-box filling machines have population weight variance 1 oz^2 each. Independent samples of size 36 have observed means $\bar{x}_A = 4.5$ oz and $\bar{x}_B = 4.7$ oz.

(a) Use the Central Limit Theorem to find $\mathbb{P}(\bar{X}_B - \bar{X}_A \geq 0.2)$ when $\mu_A = \mu_B$.

(b) Does this experiment strongly support a difference between the two population means? Explain using part (a).

Solution. Under equal means, $D = \bar{X}_B - \bar{X}_A$ has

$$\mathbb{E}[D] = 0, \qquad \mathrm{Var}(D) = \frac{1}{36} + \frac{1}{36} = \frac{1}{18}.$$

(a) By the CLT,

$$\mathbb{P}(D \geq 0.2) \approx 1 - \Phi\left(\frac{0.2}{\sqrt{1/18}}\right) = 1 - \Phi(0.8485) \approx \boxed{0.1981}.$$

(b) A difference at least this large in the observed direction occurs about 20% of the time even if the population means coincide. Thus the observed difference does not strongly support unequal means.

□

Exercise 8.33. The benzene concentration in the discharge water of a chemical process must not exceed 7950 parts per million (ppm). A random sample of 25 measurements has mean $\bar{x} = 7960$ ppm; the historical population standard deviation is $\sigma = 100$ ppm.

(a) Use the Central Limit Theorem to find the probability that the sample mean exceeds 7950 ppm if the population mean is exactly 7950 ppm.

(b) Is the observed $\bar{x} = 7960$ firm evidence that the population mean exceeds the limit? Under $\mu = 7950$, calculate $\mathbb{P}(\bar{X} \geq 7960)$.

Assume the benzene concentrations are normally distributed.

Solution. Under $\mu = 7950$, the sample mean is normally distributed with standard error

$$\sigma_{\bar{X}} = \frac{100}{\sqrt{25}} = 20 \text{ ppm}.$$

(a) By symmetry of the normal distribution,

$$\mathbb{P}(\bar{X} > 7950) = \mathbb{P}(Z > 0) = \boxed{0.5}.$$

(b) The observed excess is $10/20 = 0.5$ standard errors. Therefore

$$\mathbb{P}(\bar{X} \geq 7960) = 1 - \Phi(0.5) \approx \boxed{0.3085}.$$

This outcome is not rare under the proposed mean, so it is not firm evidence that the population mean violates the limit.

□

Exercise 8.34. Two alloys A and B have population standard deviations of load capacity equal to 5 tons each. Independent samples of 30 specimens of each alloy give $\bar{x}_A = 49.5$ and $\bar{x}_B = 45.5$ tons, a difference of 4 tons. Alloy A's manufacturers regard this as conclusive evidence that $\mu_A > \mu_B$, while alloy B's manufacturers argue that such a difference could arise by chance if the means were equal.

(a) Evaluate alloy B's argument by computing $\mathbb{P}(\bar{X}_A - \bar{X}_B > 4)$ under $\mu_A = \mu_B$.
(b) Do the data strongly support alloy A? Explain.

Solution. Under equal population means, $D = \bar{X}_A - \bar{X}_B$ has

$$\mathbb{E}[D] = 0, \qquad \sigma_D = \sqrt{\frac{5^2}{30} + \frac{5^2}{30}} = \sqrt{\frac{5}{3}} \approx 1.291.$$

(a) Applying the normal approximation,

$$\mathbb{P}(D > 4) \approx 1 - \Phi\left(\frac{4}{\sqrt{5/3}}\right) = 1 - \Phi(3.0984) \approx \boxed{0.000973}.$$

(b) Such a large positive difference is extremely unlikely under equal means. The evidence strongly supports a greater mean load capacity for alloy A, although the sample does not establish that inequality with certainty.

□

Exercise 8.36. Let $X_1, X_2, \ldots, X_n$ be an independent random sample of strictly positive random variables. Assume that $\mathbb{E}[\ln X_1]$ and $\mathrm{Var}(\ln X_1)$ are finite, with positive variance. Use the Central Limit Theorem to argue that, for sufficiently large n, the product $Y = X_1 X_2 \cdots X_n$ is approximately lognormally distributed.

Solution. Put $U_i = \ln X_i$, and write $\mu_U = \mathbb{E}[U_1]$ and $\sigma_U^2 = \mathrm{Var}(U_1)$, where $0 < \sigma_U^2 < \infty$. Since the original sample is i.i.d., so is (U_i), and

$$\ln Y = \sum_{i=1}^{n} \ln X_i = \sum_{i=1}^{n} U_i.$$

The Central Limit Theorem gives

$$\frac{\ln Y - n\mu_U}{\sigma_U \sqrt{n}} \approx \mathcal{N}(0,1) \quad \text{for large } n.$$

Thus $\ln Y$ is approximately normal with mean $n\mu_U$ and variance $n\sigma_U^2$. Since $Y = e^{\ln Y} > 0$, it follows that $\boxed{Y \text{ is approximately lognormal}}$, with log-parameters $n\mu_U$ and $n\sigma_U^2$. □

Exercise 8.37. For a chi-squared distribution, find the right-tail critical values

(a) $\chi^2_{0.005}(10)$;
(b) $\chi^2_{0.05}(6)$;
(c) $\chi^2_{0.01}(16)$.

Here $\mathbb{P}(Q > \chi^2_\alpha(\nu)) = \alpha$ for $Q \sim \chi^2_\nu$.

Solution. Read the right-tail critical values from the chi-squared table, using the degrees of freedom in parentheses:

(a) $\boxed{\chi^2_{0.005}(10) \approx 25.188}$.

(b) $\boxed{\chi^2_{0.05}(6) \approx 12.592}$.

(c) $\boxed{\chi^2_{0.01}(16) \approx 32.000}$.

□

Exercise 8.38. For a chi-squared distribution, find

(a) $\chi^2_{0.025}(12)$;
(b) $\chi^2_{0.01}(14)$;
(c) $\chi^2_{0.95}(8)$.

Use the right-tail convention $\mathbb{P}(Q > \chi^2_\alpha(\nu)) = \alpha$ for $Q \sim \chi^2_\nu$.

Solution. The right-tail areas 0.025, 0.01, and 0.95 determine the table columns:

(a) $\boxed{\chi^2_{0.025}(12) \approx 23.337}$.

(b) $\boxed{\chi^2_{0.01}(14) \approx 29.141}$.

(c) $\boxed{\chi^2_{0.95}(8) \approx 2.733}$.

□

Exercise 8.39. For $Q \sim \chi^2_\nu$, find $\chi^2_\alpha(\nu)$ satisfying

(a) $\mathbb{P}(Q > \chi^2_\alpha(5)) = 0.95$;
(b) $\mathbb{P}(Q > \chi^2_\alpha(13)) = 0.99$;
(c) $\mathbb{P}(19.68 < Q < \chi^2_\alpha(11)) = 0.04$.

Solution.

(a) The defining right-tail area is $\alpha = 0.95$, so $\boxed{\chi^2_{0.95}(5) \approx 1.145}$.

(b) Here $\alpha = 0.99$, so $\boxed{\chi^2_{0.99}(13) \approx 4.107}$.

(c) For 11 degrees of freedom the table gives $19.68 \approx \chi^2_{0.05}(11)$. Thus

$$\begin{aligned} 0.04 &= \mathbb{P}(19.68 < Q < \chi^2_\alpha(11)) \\ &\approx 0.05 - \alpha, \end{aligned}$$

so $\alpha = 0.01$ and $\boxed{\chi^2_{0.01}(11) \approx 24.725}$.

□

Exercise 8.40. For a chi-squared random variable Q with the indicated degrees of freedom, find the critical value $\chi^2_\alpha(\nu)$ satisfying

(a) $\mathbb{P}(Q < \chi^2_\alpha(7)) = 0.95$, where $Q \sim \chi^2_7$;
(b) $\mathbb{P}(4.601 < Q < \chi^2_\alpha(15)) = 0.005$, where $Q \sim \chi^2_{15}$;
(c) $\mathbb{P}(\chi^2_\alpha(19) < Q < 32.852) = 0.95$, where $Q \sim \chi^2_{19}$.

Solution.

(a) The probability to the right is $\alpha = 1 - 0.95 = 0.05$. Hence $\boxed{\chi^2_{0.05}(7) \approx 14.067}$.

(b) The chi-square table gives $4.601 \approx \chi^2_{0.995}(15)$, so

$$0.005 = \mathbb{P}(4.601 < Q < \chi^2_\alpha(15)) \approx 0.995 - \alpha.$$

Thus $\alpha = 0.99$ and $\boxed{\chi^2_{0.99}(15) \approx 5.229}$.

(c) The table gives $32.852 \approx \chi^2_{0.025}(19)$. The total left-tail probability at the lower limit must therefore be $0.975 - 0.95 = 0.025$, so its right-tail probability is $\alpha = 0.975$. Thus $\boxed{\chi^2_{0.975}(19) \approx 8.907}$.

□

Exercise 8.41. Assume sample variances are measured continuously. A random sample of 30 observations is taken from a normal population with variance $\sigma^2 = 5$. Find the probability that the sample variance S^2 is

(a) greater than 7.338;

(b) between 2.766 and 7.883.

Solution. Since the population is normal,

$$Q = \frac{(n-1)S^2}{\sigma^2} = \frac{29S^2}{5} \sim \chi^2_{29}.$$

(a) The threshold is $29(7.338)/5 \approx 42.56$, the right-tail 0.05 critical value for 29 degrees of freedom. Thus

$$\mathbb{P}(S^2 > 7.338) = \mathbb{P}(Q > 42.56) \approx \boxed{0.05}.$$

(b) The two thresholds are

$$\frac{29(2.766)}{5} \approx 16.043, \qquad \frac{29(7.883)}{5} \approx 45.721.$$

These are approximately $\chi^2_{0.975}(29)$ and $\chi^2_{0.025}(29)$, leaving 0.025 in each tail. Therefore

$$\mathbb{P}(2.766 < S^2 < 7.883) \approx \boxed{0.95}.$$

□

Exercise 8.42. Placement-test scores over the past five years are approximately normally distributed with mean 74 and variance $\sigma^2 = 8$. Would $\sigma^2 = 8$ still seem plausible if a random sample of 20 students taking the test this year has sample variance $s^2 = 20$?

Solution. Under the proposed variance $\sigma^2 = 8$ and the normal-population assumption,

$$Q = \frac{(20-1)S^2}{8} \sim \chi^2_{19}.$$

The observed statistic is $q = 19(20)/8 = 47.5$, so the probability of a sample variance at least this large is

$$\mathbb{P}(S^2 \geq 20 \mid \sigma^2 = 8) = \mathbb{P}(\chi^2_{19} \geq 47.5) \approx \boxed{0.000303}.$$

This is an exceptionally rare outcome if the variance remains 8, giving strong evidence that the old variance is no longer appropriate; it does not prove the new variance equals 20. □

Exercise 8.43. Show that the variance of the sample variance S^2 for independent random samples of size n from a normal population decreases as n increases. Assume $n \geq 2$ and population variance $\sigma^2 > 0$. *Hint.* First find the variance of $(n-1)S^2/\sigma^2$.

Solution. For a normal sample, the sample-variance sampling theorem gives

$$Q = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad \operatorname{Var}(Q) = 2(n-1).$$

Because $S^2 = \sigma^2 Q/(n-1)$,

$$\begin{aligned}\operatorname{Var}(S^2) &= \left(\frac{\sigma^2}{n-1}\right)^2 \operatorname{Var}(Q) \\ &= \frac{\sigma^4}{(n-1)^2}\, 2(n-1) = \boxed{\frac{2\sigma^4}{n-1}}.\end{aligned}$$

For fixed σ^2, this variance decreases to 0 as $n \to \infty$. □

Exercise 8.44. Find the following critical values of Student's t-distribution, where $t_\alpha(\nu)$ leaves probability α to its right:

(a) $t_{0.05}(16)$;
(b) $-t_{0.05}(12)$;
(c) $t_{0.95}(9)$.

Solution. Using the right-tail t-table and the symmetry $t_{1-\alpha}(\nu) = -t_\alpha(\nu)$,

(a) $\boxed{t_{0.05}(16) \approx 1.746}$.
(b) $\boxed{-t_{0.05}(12) \approx -1.782}$.
(c) $t_{0.95}(9) = -t_{0.05}(9)$, so $\boxed{t_{0.95}(9) \approx -1.833}$.

□

Exercise 8.45. For $T \sim t_\nu$, find

(a) $\mathbb{P}(T < 2.65)$ when $\nu = 13$;
(b) $\mathbb{P}(T > 2.06)$ when $\nu = 25$;
(c) $\mathbb{P}(-1.74 < T < 2.567)$ when $\nu = 17$;
(d) $\mathbb{P}(T > -2.06)$ when $\nu = 25$.

Solution. Using the t-table with right-tail areas and symmetry,

(a) $2.65 \approx t_{0.01}(13)$, hence $\boxed{\mathbb{P}(T < 2.65) \approx 0.99}$.
(b) $2.06 \approx t_{0.025}(25)$, hence $\boxed{\mathbb{P}(T > 2.06) \approx 0.025}$.
(c) $1.74 \approx t_{0.05}(17)$ and $2.567 \approx t_{0.01}(17)$. The omitted tails have areas 0.05 and 0.01, respectively, giving

$$\boxed{\mathbb{P}(-1.74 < T < 2.567) \approx 1 - 0.05 - 0.01 = 0.94}.$$

(d) By symmetry, $\mathbb{P}(T > -2.06) = \mathbb{P}(T < 2.06) \approx \boxed{0.975}$.

□

Exercise 8.46. For a Student's t random variable T, find

$$\mathbb{P}(-t_{0.01} < T < t_{0.05}),$$

where t_α denotes the right-tail critical value for the same degrees of freedom.

Solution. By symmetry, the area to the left of $-t_{0.01}$ is 0.01; the area to the right of $t_{0.05}$ is 0.05. Hence

$$\mathbb{P}(-t_{0.01} < T < t_{0.05}) = 1 - 0.01 - 0.05 = \boxed{0.94}.$$

□

Exercise 8.47. For a random sample of size 30 from a normal population, let T be the Student statistic. Find k such that

(a) $\mathbb{P}(-1.699 < T < k) = 0.945$;

(b) $\mathbb{P}(k < T < 2.045) = 0.025$;

(c) $\mathbb{P}(-k < T < k) = 0.98$.

Solution. Here $T \sim t_{29}$. From the table, $1.699 \approx t_{0.05}(29)$ and $2.045 \approx t_{0.025}(29)$.

(a) The area to the left of -1.699 is 0.05, so the area to the left of k is $0.05 + 0.945 = 0.995$. Thus

$$\boxed{k = t_{0.005}(29) \approx 2.756}.$$

(b) The area to the left of 2.045 is 0.975. Hence the area to the left of k must be $0.975 - 0.025 = 0.95$, and

$$\boxed{k = t_{0.05}(29) \approx 1.699}.$$

(c) The interval is symmetric and has two tails of size 0.01, giving

$$\boxed{k = t_{0.01}(29) \approx 2.462}.$$

□

Exercise 8.48. A technical training firm claims a mean training time of 40 hours. It checks 24 trainees monthly and accepts the claim if the computed Student statistic lies between $-t_{0.025}$ and $t_{0.025}$. The observed mean and standard deviation are $\bar{x} = 42$ hours and $s = 6$ hours. What conclusion should the firm draw? Assume training times are approximately normally distributed.

Solution. Using the claimed $\mu = 40$, the observed Student statistic and its degrees of freedom are

$$t = \frac{42 - 40}{6/\sqrt{24}} \approx 1.633, \qquad \nu = 23.$$

The t-table gives $t_{0.025}(23) \approx 2.069$. Since

$$-2.069 < 1.633 < 2.069,$$

the statistic lies inside the prescribed acceptance interval. The sample is compatible with the 40-hour claim, so the firm has no reason, under its stated rule, to change the claimed mean. □

Exercise 8.49. A normal population has mean 15 and unknown variance. Is it plausible to obtain a random sample of size 16 with observed mean 18 and sample standard deviation 3.8? If not, what conclusion would you draw?

Solution. Under the proposed population mean $\mu = 15$,

$$t = \frac{18 - 15}{3.8/\sqrt{16}} = \frac{3}{0.95} \approx 3.158, \qquad \nu = 15.$$

The observed value exceeds $t_{0.005}(15) \approx 2.947$, so it lies well into the upper tail. The two-sided probability of a deviation at least this extreme is

$$2\mathbb{P}(t_{15} > 3.158) \approx \boxed{0.0065}.$$

Thus this sample would be unusual under $\mu = 15$, giving evidence that the population mean is different from 15 (in the direction of a larger mean). □

Exercise 8.50. A transport company claims that the mean bus running time on a particular route is 300 minutes. The running times of six randomly observed buses are

$$320,\ 310,\ 295,\ 312,\ 302,\ 308 \text{ minutes.}$$

Would you agree with the company's claim? Assume the running-time population is normal.

Solution. The sample statistics are

$$\bar{x} = \frac{1847}{6} \approx 307.833, \qquad s^2 = \frac{1}{5}\sum_{i=1}^{6}(x_i - \bar{x})^2 \approx 73.7667, \quad s \approx 8.589.$$

Under $\mu = 300$,

$$t = \frac{\bar{x} - 300}{s/\sqrt{6}} \approx 2.234, \qquad \nu = 5.$$

Since $t_{0.025}(5) \approx 2.571$, the observed statistic lies within the central 95% interval. The two-sided tail probability is about 0.0758. The data do not provide strong evidence against the company's 300-minute claim at the usual 5% level, though the sample mean is higher. □

Exercise 8.51. For an F-distribution, find the right-tail critical values

(a) $f_{0.01}(12, 10)$;
(b) $f_{0.01}(10, 12)$;
(c) $f_{0.05}(15, 8)$;
(d) $f_{0.95}(10, 5)$;
(e) $f_{0.99}(18, 14)$.

The first and second degrees of freedom refer to the numerator and denominator, respectively.

Solution. Use Table A.6 for upper-tail values and the reciprocal identity

$$f_{1-\alpha}(\nu_1, \nu_2) = \frac{1}{f_{\alpha}(\nu_2, \nu_1)}.$$

(a) $\boxed{f_{0.01}(12, 10) \approx 4.71}$.
(b) $\boxed{f_{0.01}(10, 12) \approx 4.30}$.
(c) $\boxed{f_{0.05}(15, 8) \approx 3.22}$.
(d) $f_{0.95}(10, 5) = 1/f_{0.05}(5, 10)$, hence $\boxed{f_{0.95}(10, 5) \approx 0.301}$.
(e) $f_{0.99}(18, 14) = 1/f_{0.01}(14, 18)$, hence $\boxed{f_{0.99}(18, 14) \approx 0.306}$.

□

Exercise 8.52. Pull-strength tests on 10 soldered leads for a semiconductor device give the following force measurements (lb) before encapsulation:

19.8	12.7	13.2	16.9	10.6
18.8	11.1	14.3	17.0	12.5

A second set of 8 leads tested after encapsulation gives

24.9	22.8	23.6	22.1
20.4	21.6	21.8	22.5

Comment on the evidence concerning equality of the population variances. For the F-distribution comparison, assume independent samples from normal populations.

Solution. Let S_1^2 be the pre-encapsulation sample variance and S_2^2 the post-encapsulation variance. From the listed measurements,

	Before	After
n	10	8
$\bar{x}$	14.6900	22.4625
s^2	10.4410	1.8455

Under equal population variances and the normality assumption,

$$F = \frac{S_1^2}{S_2^2} \sim F_{9,7}, \qquad f_{\text{obs}} = \frac{10.4410}{1.8455} \approx 5.6574.$$

The table gives $f_{0.05}(9,7) \approx 3.68$ and $f_{0.01}(9,7) \approx 6.72$, so the observed ratio is unusually large. The upper-tail probability is approximately 0.0162 (and the two-sided probability about 0.0324). There is evidence against equal population variances; the post-encapsulation sample shows less variability. This comparison does not by itself establish a causal reduction in variance. □

Exercise 8.53. The heat-producing capacities of coal from two mines (in millions of calories per ton) were measured as follows:

Mine 1	8260	8130	8350	8070	8340	—
Mine 2	7950	7890	7900	8140	7920	7840

Can the two population variances reasonably be regarded as equal? For the F-distribution comparison, assume independent samples from normal populations.

Solution. The sample sizes and variances obtained from the data are

	Mine 1	Mine 2
n	5	6
$\bar{x}$	8230	7940
s^2	15750	10920

Under the equal-variance model,

$$F = \frac{S_1^2}{S_2^2} \sim F_{4,5}, \qquad f_{\text{obs}} = \frac{15750}{10920} \approx 1.4423.$$

This is far below $f_{0.05}(4,5) \approx 5.19$; its upper-tail probability is about 0.3436. Therefore the observed difference in sample variances is quite plausible under equal population variances. The data give no strong evidence against equality, although they cannot establish exact equality. □

Exercise 8.64. Independent random samples of sizes $n_1 = 16$ and $n_2 = 27$ are taken from normal populations with variances $\sigma_1^2 = 8$ and $\sigma_2^2 = 14$, respectively. Find

$$\mathbb{P}\left(\frac{S_1^2}{S_2^2} > 1.873\right).$$

Solution. For independent normal samples, the scaled variance ratio satisfies

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{14}{8}\frac{S_1^2}{S_2^2} \sim F_{15,26}.$$

Therefore

$$\begin{aligned}\mathbb{P}\left(\frac{S_1^2}{S_2^2} > 1.873\right) &= \mathbb{P}\left(F_{15,26} > 1.873 \cdot \frac{14}{8}\right) \\ &= \mathbb{P}(F_{15,26} > 3.27775) \approx \boxed{0.00388}.\end{aligned}$$

In particular, the F-table has $f_{0.01}(15, 26) \approx 2.815 < 3.27775$, confirming that this probability is below 0.01. □

Exercise 8.67. The breaking strength X of a type of rivet has mean 5000 psi and standard deviation 400 psi. Consider the mean $\bar{X}$ of an independent random sample.

(a) For a sample of size 36, find $\mathbb{P}(4800 < \bar{X} < 5200)$.

(b) Using the normal approximation, find the smallest integer sample size n for which $\mathbb{P}(4900 < \bar{X} < 5100)$ is at least 0.99.

Solution.

(a) For $n = 36$, the standard error is $400/\sqrt{36} = 400/6$. By the CLT,

$$\begin{aligned}\mathbb{P}(4800 < \bar{X} < 5200) &\approx \mathbb{P}(-3 < Z < 3) = 2\Phi(3) - 1 \\ &\approx \boxed{0.9973}.\end{aligned}$$

(b) For general n, the CLT gives

$$\mathbb{P}(4900 < \bar{X} < 5100) \approx 2\Phi\left(\frac{100\sqrt{n}}{400}\right) - 1.$$

The normal approximation reaches at least 0.99 when

$$\frac{\sqrt{n}}{4} \geq \Phi^{-1}(0.995) \approx 2.575829, \quad \text{so} \quad n \geq (4 \cdot 2.575829)^2 \approx 106.158.$$

Hence the minimum integer sample size under this approximation is $\boxed{n = 107}$.

□